# 2026 Senate forecast - Most_Magnificent_Model scheduled

Team 5: Elizabeth Miller, Mark Scott, Ethan Verrone, Boaz Alkalay, Leo Westebbe

**Code and data:** GitHub repository (https://github.com/MarkGScott960/2026_Midterm_Forecast) | This model's folder (https://github.com/MarkGScott960/2026_Midterm_Forecast/tree/main/Senate%20Forecast/Final%20Model)

scheduled run, 2026-11-01 11:10

P(DEMOCRATS >= 51 SEATS)

**77.8%**

simulation precision ±0.3 pts

EXPECTED DEMOCRATIC SEATS

**52.4**

median 52 | 80%: 49-55 | 95%: 47-57

KALSHI CONTROL MARKET

**71.5%**

sum of race prices: 52.3 seats

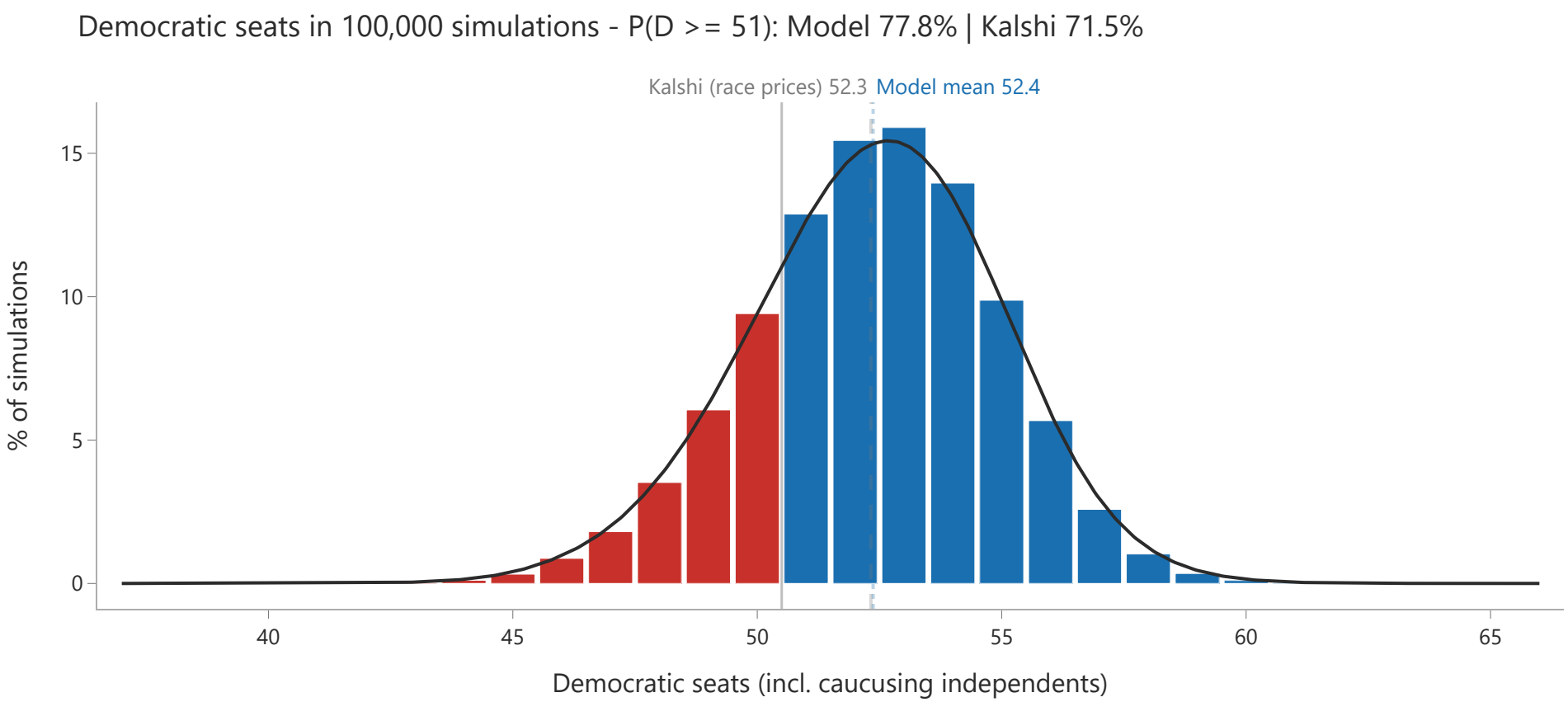
RUN

**2026-11-01 11:10**

100,000 simulations | data as of 2026-11-01 | 5 min

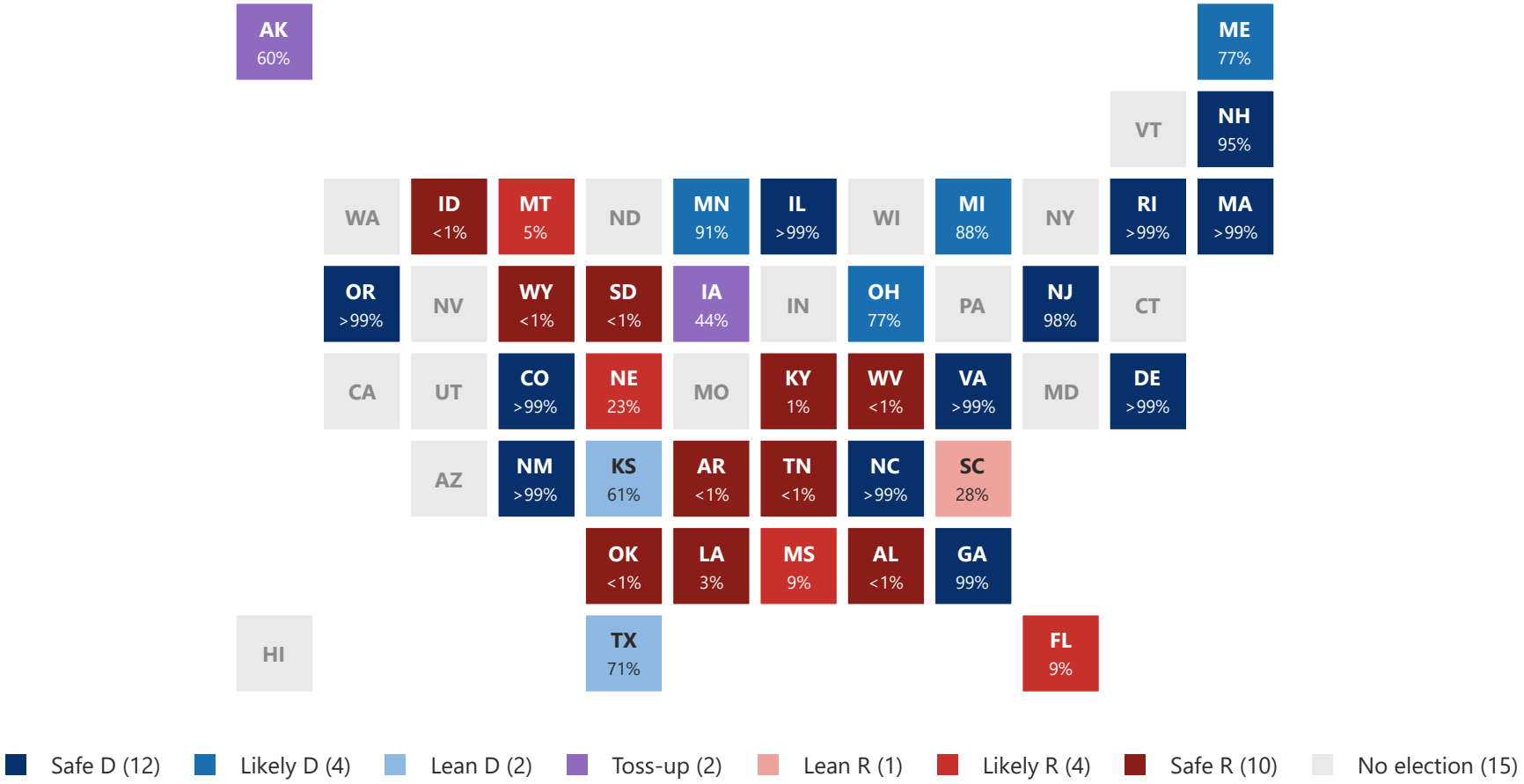
The model and Kalshi expect about the same number of Democratic seats (52.4 vs 52.3); the model's higher control odds come from rating Michigan, Kansas, Maine and Ohio higher for Democrats than the market does.

**Seat distribution**

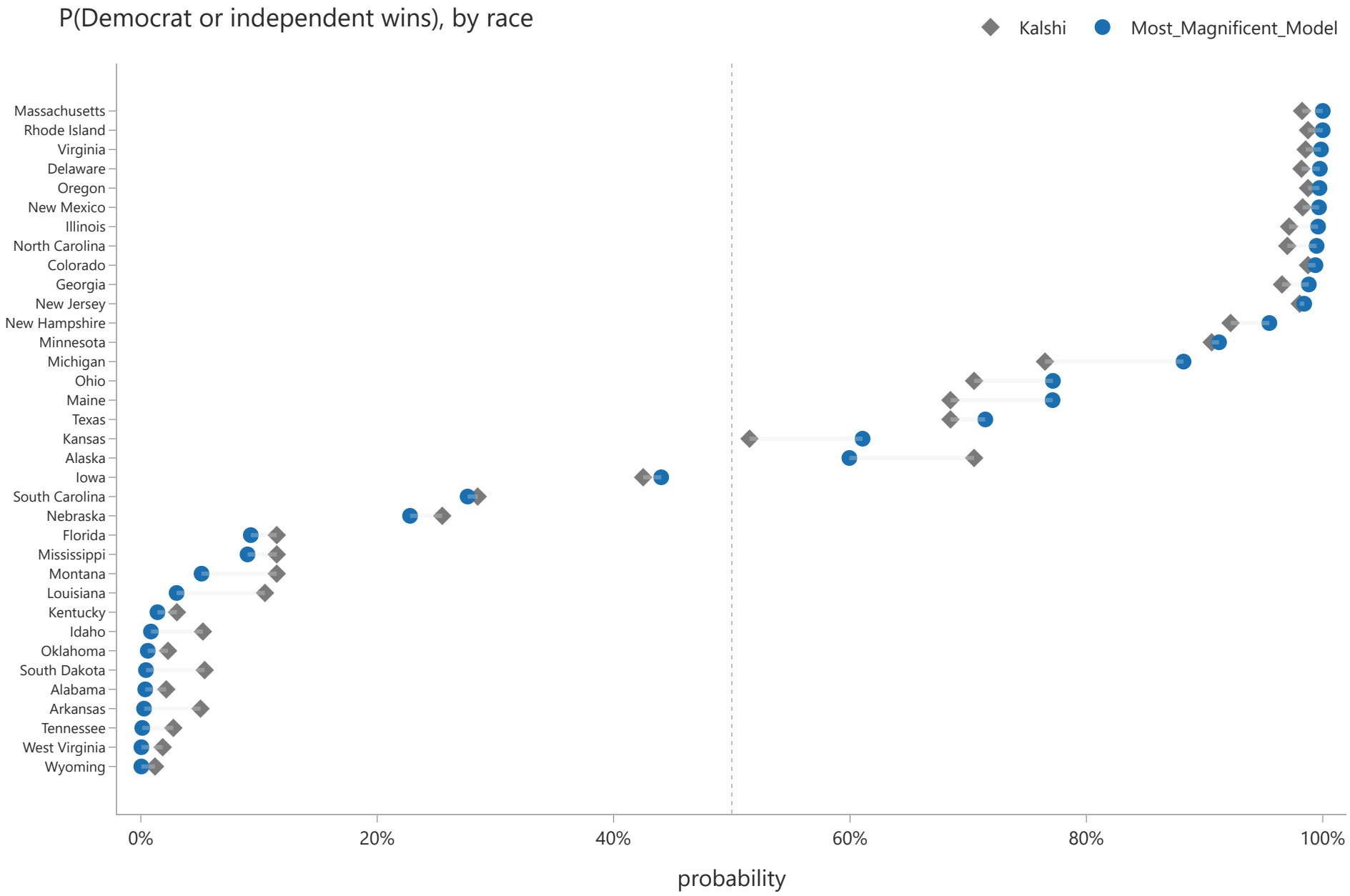


## Map

Race ratings (number = Model P(D/Ind wins); D includes independents)
Toss-up under 60%, Lean 60%-75%, Likely 75%-95%, Safe 95%+



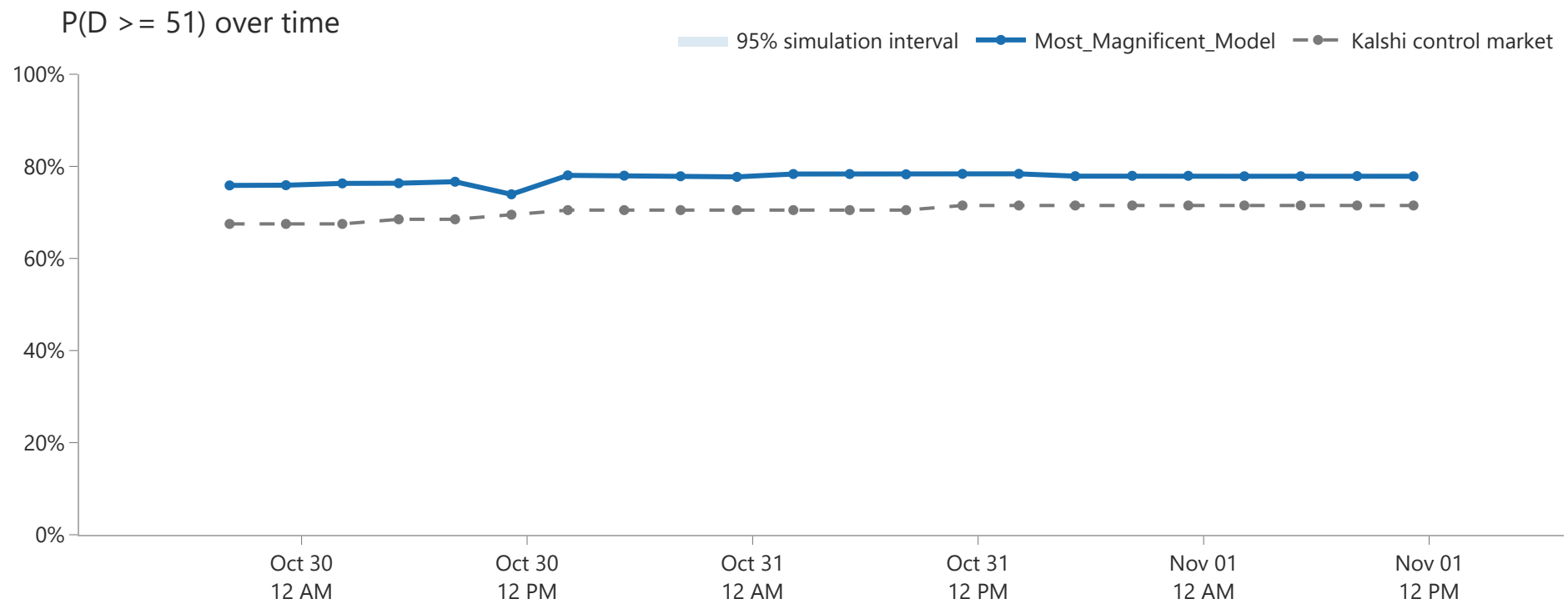
## Race by race

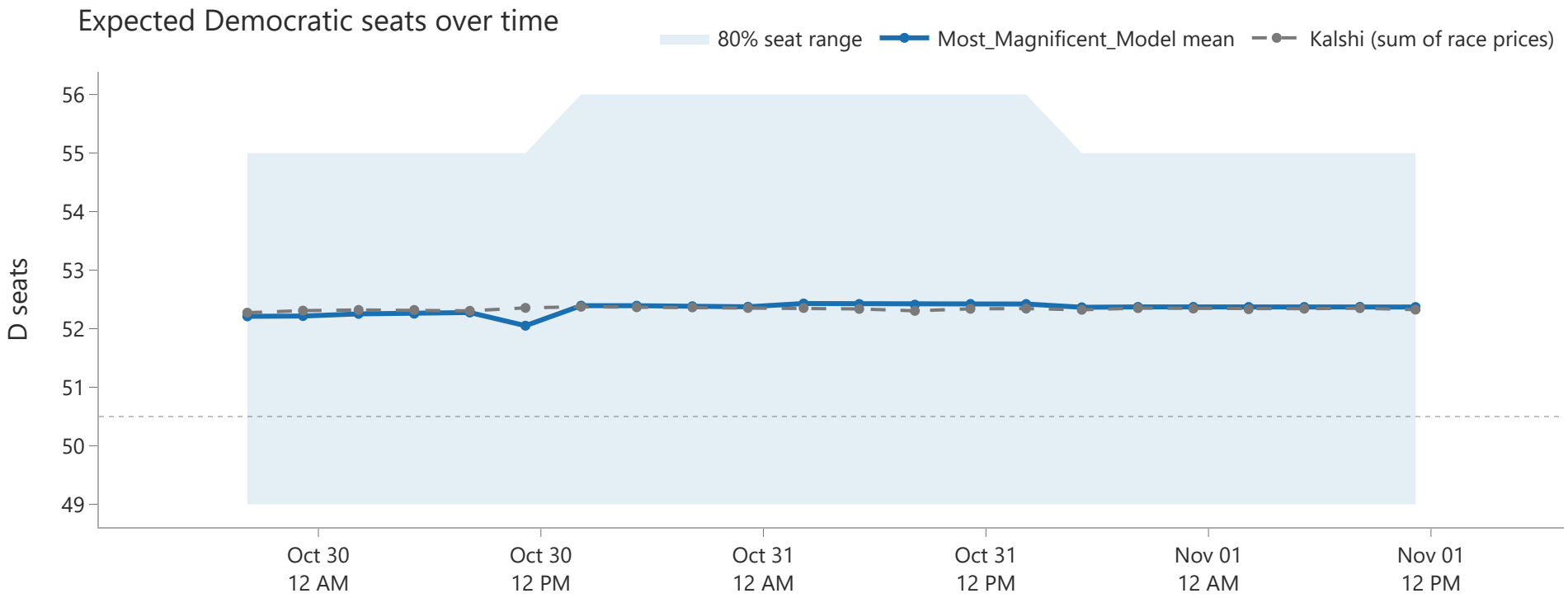

P(Democrat or independent wins), by race
Kalshi
Most_Magnificent_Model
Massachusetts
Rhode Island
Virginia
Delaware
Oregon
New Mexico
Illinois
North Carolina
Colorado
Georgia
New Jersey
New Hampshire
Minnesota
Michigan
Ohio
Maine
Texas
Kansas
Alaska
Iowa
South Carolina
Nebraska
Florida
Mississippi
Montana
Louisiana
Kentucky
Idaho
Oklahoma
South Dakota
Alabama
Arkansas
Tennessee
West Virginia
Wyoming
0%
20%
40%
60%
80%
100%
probability

| Race | Call | Model P(D/Ind) | Kalshi P(D/Ind) | Model - Kalshi (pts) | Projected margin | +/- (1 SD) | Polls | Model change since first run (pts) | Kalshi change since first run (pts) |
|---|---|---|---|---|---|---|---|---|---|
| Massachusetts | Safe D | >99% | 98% | +2 | D+34.2 | +/-5.6 | 13 | +0 | -0 |
| Rhode Island | Safe D | >99% | 99% | +1 | D+25.4 | +/-5.4 | 4 | +0 | -0 |
| Virginia | Safe D | >99% | 99% | +1 | D+18.8 | +/-5.8 | 1 | +0 | +0 |
| Delaware | Safe D | >99% | 98% | +2 | D+28.7 | +/-9.4 | 0 | +0 | -0 |
| Oregon | Safe D | >99% | 99% | +1 | D+28.7 | +/-9.4 | 0 | -0 | -0 |
| New Mexico | Safe D | >99% | 98% | +1 | D+18.8 | +/-6.4 | 2 | +0 | -0 |
| Illinois | Safe D | >99% | 97% | +2 | D+14.7 | +/-5.3 | 3 | -0 | -1 |
| North Carolina | Safe D | >99% | 97% | +2 | D+12.3 | +/-4.6 | 40 | +0 | +1 |
| Colorado | Safe D | >99% | 99% | +1 | D+24.7 | +/-9.4 | 0 | +0 | +1 |
| Georgia | Safe D | 99% | 97% | +2 | D+11.0 | +/-4.7 | 18 | +0 | +0 |
| New Jersey | Safe D | 98% | 98% | +0 | D+20.7 | +/-9.4 | 0 | -0 | +1 |
| New Hampshire | Safe D | 95% | 92% | +3 | D+8.3 | +/-4.9 | 21 | +1 | +1 |
| Minnesota | Likely D | 91% | 91% | +1 | D+6.3 | +/-4.8 | 7 | +1 | +1 |
| Michigan | Likely D | 88% | 76% | +12 | D+5.3 | +/-4.6 | 43 | +1 | +2 |
| Ohio | Likely D | 77% | 70% | +7 | D+3.3 | +/-4.5 | 32 | +1 | +1 |
| Maine | Likely D | 77% | 68% | +9 | D+3.2 | +/-4.5 | 18 | +2 | +5 |
| Texas | Lean D | 71% | 68% | +3 | D+2.4 | +/-4.3 | 53 | +2 | +0 |
| Kansas | Lean D | 61% | 52% | +10 | D+1.2 | +/-4.5 | 7 | +13 | +3 |
| Alaska | Toss-up | 60% | 70% | -11 | D+1.1 | +/-4.6 | 18 | +0 | +2 |
| Iowa | Toss-up | 44% | 42% | +2 | R+0.7 | +/-4.6 | 24 | -1 | -5 |
| South Carolina | Lean R | 28% | 28% | -1 | R+2.6 | +/-4.6 | 4 | -1 | +1 |
| Nebraska | Likely R | 23% | 26% | -3 | R+3.6 | +/-4.9 | 4 | -3 | -7 |

| Race | Call | Model P(D/Ind) | Kalshi P(D/Ind) | Model - Kalshi (pts) | Projected margin | +/- (1 SD) | Polls | Model change since first run (pts) | Kalshi change since first run (pts) |
|---|---|---|---|---|---|---|---|---|---|
| Florida | Likely R | 9% | 11% | -2 | R+5.8 | +/-4.5 | 17 | -1 | +1 |
| Mississippi | Likely R | 9% | 11% | -2 | R+7.1 | +/-5.4 | 3 | -0 | +0 |
| Montana | Likely R | 5% | 11% | -6 | R+9.3 | +/-5.7 | 26 | +1 | -1 |
| Louisiana | Safe R | 3% | 10% | -7 | R+13.2 | +/-7.0 | 0 | +0 | +1 |
| Kentucky | Safe R | 1% | 3% | -2 | R+21.1 | +/-9.4 | 0 | -0 | -0 |
| Idaho | Safe R | <1% | 5% | -4 | R+17.2 | +/-7.0 | 1 | -1 | -0 |
| Oklahoma | Safe R | <1% | 2% | -2 | R+25.1 | +/-9.4 | 0 | -0 | -0 |
| South Dakota | Safe R | <1% | 5% | -5 | R+13.9 | +/-5.1 | 4 | +0 | -0 |
| Alabama | Safe R | <1% | 2% | -2 | R+16.2 | +/-5.7 | 5 | -0 | +0 |
| Arkansas | Safe R | <1% | 5% | -5 | R+14.5 | +/-4.9 | 5 | -0 | +0 |
| Tennessee | Safe R | <1% | 3% | -3 | R+22.9 | +/-6.8 | 1 | -0 | -1 |
| West Virginia | Safe R | <1% | 2% | -2 | R+36.9 | +/-9.4 | 0 | -0 | -0 |
| Wyoming | Safe R | <1% | 1% | -1 | R+37.1 | +/-9.4 | 0 | -0 | +0 |

## History






## Notes

- Model Testing: 35 PASS | 6 WARN | 0 FAIL (details in model_warnings.xlsx).
- Data: votehub_senate live, votehub_generic_ballot live, votehub_approval live, kalshi_races live, kalshi_national live.

## Key settings

| | |
|---|---|
| Poll source | VoteHub API |
| Poll age | all polls, half-life 7 days |
| Polls : fundamentals : Kalshi | Bayesian precision (poll floor 3.9, prior SD 9.3); Kalshi precision 1/8^2, longshot-corrected, conversion SD >= 4.42 |
| Poll reliability (thin races) | automatic (Bayesian precision) |
| Fundamentals | PVI x 2 + national environment (generic-ballot blend 0.75) + candidate effects x 0.30 |
| Incumbency | 3.8 pts (appointed 1) |
| LV : RV : A | 1 : 0.85 : 0.7; RV shift +0.9 (measured) |
| Flooding penalty | 1/k^0.5 |
| House effects | in-cycle, shrunk to scorecard (k 5) |
| Ungraded pollsters | random |
| National signals | VoteHub polls + Kalshi averages (= 5 polls) |
| Kalshi races | Bayesian input + comparison (found automatically) |
| Source errors | correlated (rho 0.3); sims: national 2.5 + region 1.5 + state, in pts |
| Calibration | calibration.json (2026-10-29 22:45) |
| Simulations | 100,000 |

## Equations and weights

1. WEIGHT OF EACH POLL

$$w = 0.5^{\text{age}/7} \times \left(\frac{\min(n,\,2500)}{800}\right)^{0.5} \times w_{\text{pop}} \times w_{\text{partisan}} \times w_{\text{generic}} \times w_{\text{field}} \times w_{\text{quality}} \times w_{\text{accuracy}} \times \frac{1}{k^{0.5}} \times \frac{1}{\text{versions}}$$

Population: LV 1, RV 0.85, adults 0.7. Partisan sponsor 0.6. Generic-party question 0.5. Full field 0.9. Quality 0.5 to 1 by VoteHub grade (ungraded firms get a random graded firm's grade in each simulation). Accuracy = (1.25 x 4 / poll SD)^2, kept between 0.5 and 2. k = the pollster's polls of that race.

2. EACH POLL'S MARGIN

$$m = (D - R)\,\frac{D + R + U}{D + R} + s_{RV} \pm s_{\text{partisan}} - h, \qquad h = \frac{\sum \text{residuals} + 5\,h_{\text{scorecard}}}{n_{\text{polls}} + 5}$$

U = undecided. RV shift s_RV = +0.88 pts (registered-voter polls only; measured from LV/RV pairs). Partisan shift = 3 pts against the sponsor. Residual = poll minus the same state's average of other pollsters, from the last 60 days.

3. POLLING AVERAGE

$$\bar{m} = \frac{\sum w\,m}{\sum w}, \qquad SD_{\text{polls}} = \sqrt{\frac{SD_{\text{poll}}^2}{n_{\text{eff}}} + 3.87^2 + SD_U^2}, \qquad n_{\text{eff}} = \frac{(\sum w)^2}{\sum w^2}$$

The middle term is the error all polls of a state share, so more polls can't remove it.

---

4. NATIONAL ENVIRONMENT

$$\text{env} = 0.75 \times \text{GB} + 0.25 \times 0.35 \times (-\text{net approval}), \qquad \text{signal} = \frac{n_{\text{fresh}} \cdot \text{VoteHub} + 5 \cdot \text{Kalshi}}{n_{\text{fresh}} + 5}$$

GB = generic ballot. Each VoteHub poll counts 0.5^(age / 14) toward n_fresh. This run: env = D+8.9.

---

5. PRIOR (FUNDAMENTALS)

$$\text{prior} = 2 \times \text{PVI} + \text{env} \pm 3.78\,[\text{incumbent}] + 0.30 \times c, \qquad SD_{\text{prior}} = 9.32$$

Appointed incumbents get 1 pt. c = candidate effect: the nominees' past margin minus a generic candidate's (capped at +/- 25). Independent-led races: SD x 1.3.

---

6. KALSHI

$$P' = \text{logistic}\,(1.15 \cdot \text{logit}\,P), \qquad \text{margin}_K = \Phi^{-1}(P') \cdot \max\,(4.42,\ SD_{F+P}), \qquad SD_K = 8$$

Used only when 6% <= P <= 94%.

---

7. EACH RACE

$$\text{margin} = \frac{\pi_F \cdot \text{prior} + \pi_P \cdot \bar{m} + \pi_K \cdot \text{margin}_K}{\pi_F + \pi_P + \pi_K}, \quad \pi = \frac{1}{SD^2}, \quad SD^2 = \frac{1}{\Pi} + \frac{2(0.3)}{\Pi^2}\left(\sqrt{\pi_F\pi_P} + \sqrt{\pi_F\pi_K} + \sqrt{\pi_P\pi_K}\right) + 0.76^2 \cdot \text{days left}$$

Pi = the sum of the three precisions. The second term widens the range because the three sources tend to miss together.

---

8. SIMULATIONS

$$\tilde{m}_i = \text{margin}_i + N + R_{g(i)} + L\,f_i + e_i, \qquad \text{D seats} = 34 + \sum_i \mathbf{1}[\tilde{m}_i > 0], \qquad P(D \geq 51) = \text{share of simulations}$$

N = national miss (SD 2.52, t with df 8). R = regional miss (SD 1.51). L = lean factor (SD 1), f_i = PVI / 10. e = the rest of the race's SD (t with df 10.9). An independent's win counts for Democrats with probability: Nebraska 0.8.

---